



RS003 Week 1 — English Worksheet

Topic: Open a Pygame Window · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This week you open your very first **Pygame window** — a blank screen your game will live in. You learn that the screen is a space measured in **width** and **height**, fill it with a deep-space colour, and keep it open with a **game loop**.

Words to know: **import, window, screen, fill, flip, game loop**

1 · Predict 🧠

Read each piece of code. Before you run it, write what you think will happen.

```
screen = pygame.display.set_mode((800, 600))
```

What does **(800, 600)** mean? Which number is across, which is down?


```
SPACE = (10, 12, 40)
screen.fill(SPACE)
```

The screen gets filled with a colour. But will you actually SEE it yet? What is still missing?


```
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
```

What is this loop watching for? What makes the window finally close?

2 · Run It 🏃

🎯 **Type the example, run it, and report what you see**

Type this into your code editor exactly as it is, then run it.

```
import pygame

pygame.init()
screen = pygame.display.set_mode((800, 600))
pygame.display.set_caption("Space Shooter")

SPACE = (10, 12, 40)

running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
    screen.fill(SPACE)
    pygame.display.flip()

pygame.quit()
```

What colour is the window? What words are in the title bar at the top?

Try clicking the X button. What happens? Which line of code made that work?

3 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Read what it is **supposed** to do, fix it, then explain why the original was wrong.

Bug A — This should open a window, but nothing appears. The screen is filled, but it never shows up.

```
import pygame
pygame.init()
screen = pygame.display.set_mode((800, 600))
screen.fill((10, 12, 40))
```

Hint: filling the screen is not enough. One line makes it visible.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This is supposed to make an **800 wide, 600 tall** window. Right now it is tall and thin instead of wide.

```
screen = pygame.display.set_mode((600, 800))
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The window opens but freezes and you cannot close it with the X button.

```
import pygame
pygame.init()
screen = pygame.display.set_mode((800, 600))
screen.fill((10, 12, 40))
pygame.display.flip()

while True:
    pass
```

Hint: the loop never listens for the QUIT event.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Modify It

Change the colour and the title to make the window yours

Go back to your working window from Part 2. Make these small changes and run after each one:

- Change `SPACE` to a colour you choose. Remember each number is `(red, green, blue)` from 0 to 255.
- Change the title text in `set_caption(...)` to your own game name.

Write down the three numbers you used and the colour they made:

Find the centre of the screen

The window is 800 across and 600 down. The exact middle is halfway across and halfway down.

Work out the centre point with maths, then write the (X, Y) for the centre:

Hint: halfway across 800 is one number, halfway down 600 is another. The centre is `(across ÷ 2, down ÷ 2)` .

5 · Make It 📷

Build your own game window and show it off

Build a window that:

1. opens at 800×600 ,
2. is filled with your own deep-space colour,
3. has your own game name in the title bar,
4. closes when you click the X button.

When it works, send a **photo or video** of your window, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I set the window size to ...

I chose the colour ... because ...

The line that makes the colour appear is ...

The game loop keeps the window open by ...

One tricky moment was when ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while your window opens and closes. Talk through it like you are teaching someone who has never seen Pygame. Try to use these words: **window**, **screen**, **fill**, **flip**, **game loop**.

1. Show your window opening with its colour and title.
2. Read the `set_mode` line out loud and say what the two numbers mean.
3. Point at the line that makes the colour appear.
4. Click the X and say which code let the window close.

Write what you will say in your video. Plan it here before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your photo or video to teacher on KakaoTalk.